భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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Towards Robust Video Anomaly Detection Using a Multi-Task Learning Approach

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Introduction & Motivation



In today's world, ensuring safety and privacy is essential.Anomaly detection in videos refers to the identification of events or behaviors that deviate from what is considered normal or expected. Detecting anomalies in video is vital for ensuring security in various domains, such as public surveillance, transportation, and healthcare, as these unexpected events can indicate potential threats or emergencies. **Video anomaly detection** focuses on identifying events that diverge from typical or expected patterns of behavior. This task is critical for ensuring security in a variety of domains, including public surveillance, transportation, and healthcare. Unexpected events in these contexts often signal potential threats or emergencies, such as accidents, criminal activities, or medical incidents. Early and accurate detection of such anomalies can help mitigate risks, prevent escalations, and ensure the well-being of individuals and communities. Despite its importance, anomaly detection remains a **highly challenging problem.** The primary reason for this is the nature of abnormal events, which are often unbounded and diverse.[1]

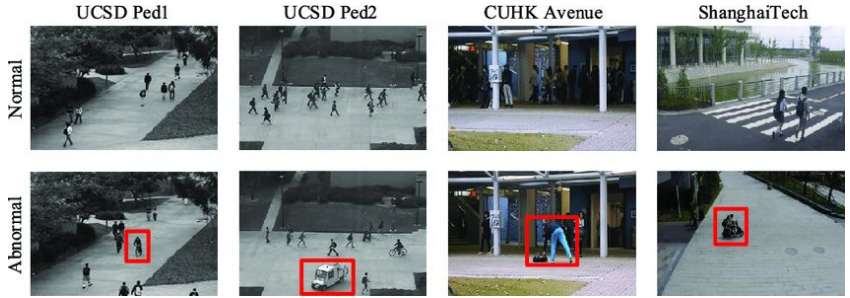


Fig 1 : Detection of Anomaly in video frames

Image Reference: Qiang, Yong, Shumin Fei, and Yiping Jiao. "Anomaly detection based on latent feature training in surveillance scenarios." *IEEE Access* 9 (2021): 68108-68117.

Problem Statement



Unlike normal events, which typically follow regular patterns and can be easily characterized, abnormal events are rare, unpredictable, and can vary widely depending on the context. For instance, an anomaly in a pedestrian crossing may involve a person running, whereas an anomaly in a factory setting may involve a machine malfunctioning. This variability makes it extremely difficult to model anomalies explicitly. Anomaly detection in videos still faces significant challenges, such as the spatiotemporal complexity of video data, the lack of labeled anomalies, and the need for models that can generalize to new types of anomalies. These challenges make it a compelling area of research, with ongoing efforts to develop more robust and scalable solutions.

Literature Review



- In the case of video anomaly detection, the main problem is the event will **occur rarely** which is unpredictable for human to predict. [1]
- Anomaly detection is **scene-dependent** which makes it to predict as it changes based on context at a particular time.[2]
- traditional methods relying on **handcrafted features**(motion patterns,appearance based,etc…), which struggled with generalization across diverse scenarios.[3]
- Traditional methods often rely on **labeled datasets**, which can be difficult to obtain, and may not effectively integrate both spatial and temporal dynamics, leading to limitations in identifying subtle anomalies.[4]
- The data-driven techniques like **deep learning** make it hard to reconstruct the normal routine events that occur rarely.
- Researchers also worked on this problem through **unsupervised learning** can be used but they lack adequate and flexible generalization of normal events. [5]
- The **pseudo-anomalous data** generation during training may not accurately capture the complexities of real-world anomalies, which could affect the model's generalization.[6]
- **Generative adversarial networks** (GANs), sought to create synthetic anomalies, yet they still faced challenges in accurately modeling real-world scenarios.[7]
- **Contrastive learning** based approach is particularly effective for learning robust representations without labeled data, making it ideal for tasks like anomaly detection,where it helps capture subtle differences by focusing on the intrinsic features of the data.[8]
- Mostly traditional approaches use a **single static network architecture** to detect anomalies, making it difficult to capture various complex patterns.[9]

Literature Review



Method	Limitation / Advantage
Scene-Dependent Models	Anomaly detection depends on scene context, which changes over time and makes predictions difficult.[2]
Traditional Handcrafted Features based models	Struggle with generalization across diverse scenarios due to reliance on motion patterns or appearance.[3]
Label-Dependent Traditional Models	Dependence on labeled datasets, lack integration of spatial-temporal dynamics, missing subtle anomalies.[4]
Data-Driven Techniques	Struggle to reconstruct rare normal events, limiting effectiveness in anomaly detection.[5]
Unsupervised Learning	Lacks adequate and flexible generalization of normal events, reducing anomaly detection accuracy.[6]
Pseudo-Anomalous Data	Synthetic anomalies may not capture real-world complexities, affecting model generalization.[7]
Generative Adversarial Networks (GAN)	Challenges in accurately modeling real-world scenarios, despite creating synthetic anomalies.[8]
Contrastive Learning	Effective for robust representation but requires careful feature design to capture subtle data differences[9]
Frame prediction	The prediction of future frames works well for detecting anomalies but struggles with extreme complexities in videos where anomalies may not deviate much from normal data.[10]
Static Network Architectures	Single architectures fail to capture complex and varied anomaly patterns in dynamic scenarios.[11]]

Implementation



To address these challenges, **multi-task learning approach** was chosen for anomaly detection.

- Frame Prediction,Self Supervised Learning and Contrastive Learning tasks were considered as in anomaly detection these approaches were best rather than labelled based methods in real-time applications.
- **Frame prediction** directly incorporates temporal consistency, which traditional methods often lack, providing an essential component for holistic video anomaly detection.
- **Self Supervised Learning** (SSL) leverages unlabeled data by designing pretext tasks, enabling the model to learn meaningful feature representations without requiring extensive labeled datasets. This is particularly advantageous for anomaly detection, where anomalies are rare and manually labeling every possible scenario is impractical.
- **Contrastive learning** plays a vital role in our framework due to its ability to learn discriminative representations. It maximizes similarity between positive pairs (e.g., augmentations of the same video segment) and minimizes similarity between negative pairs (e.g., different segments).

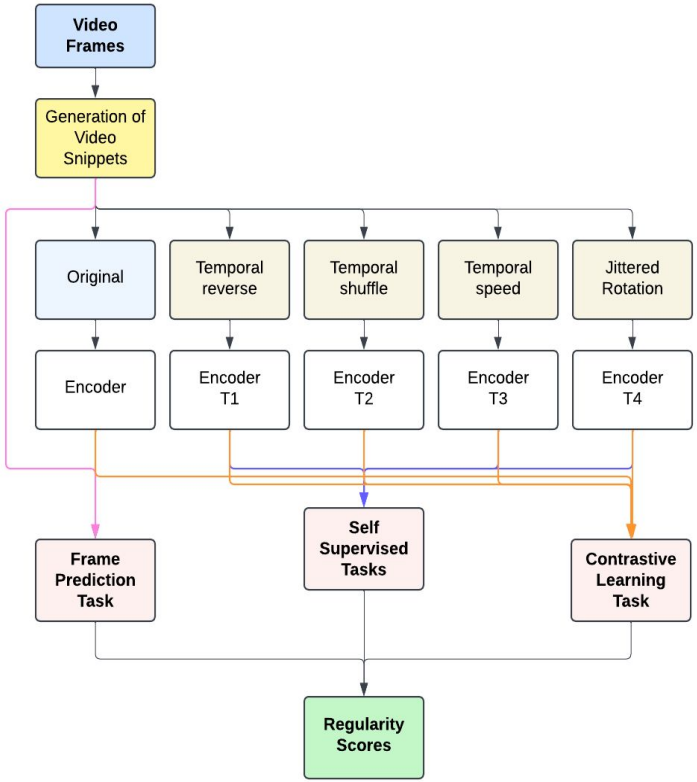


Fig 2 : Block Diagram of proposed model

Implementation



Datasets :

UCSD Ped1 was chosen in which it contains Videos of pedestrians walking in a walkway area and Anomalies include non-pedestrian entities like bicycles, vehicles, or people performing unusual actions.14,000 frames, which are divided into 34(6800 frames) and 36 videos(7200 frames) for training and testing, respectively. There are 12 types of abnormal events, including: carts, cars, skateboarding, and cycling among pedestrians.Videos are grayscale with a resolution of **158 × 238 pixels** per frame.

Train-Test Split:

Training set: Contains only normal behaviors.

Testing set: Contains both normal and anomalous behaviors.

Given a frame $I_i \in R^{H \times W \times 3}$, a snippet S_i is formed by stacking in consecutive frames with non-overlapping between snippets.

Reference :Li, Weixin, Vijay Mahadevan, and Nuno Vasconcelos. "Anomaly detection and localization in crowded scenes." *IEEE transactions on pattern analysis and machine intelligence* 36.1 (2013): 18-32.

Implementation



Data Augmentation :

We designed four temporal transformations, including temporal reverse, jittered rotation, speed, and shuffle i.e, a set of temporal transformations T={T1,T2,…,T4}to the snippet, generating augmented snippets $\{S_i^{T1}, S_i^{T2}, \ldots, S_i^{T4}\}$.

- The **Temporal reverse task** takes temporal direction of video snippets as self supervision signals. Specifically, it tries to discriminate whether the temporal order of a snippet is forward or backward.
- Four types of **jittered rotations**, i.e.,0◦,90◦, 180◦, and 270◦ for every frame same like fig ,on the raw snippet were applied as shown in Fig 3. Besides, a random noise ∈ [−3◦,3◦] is added to the original rotation.
- For the Given a video snippet, we randomly **shuffle** its temporal orders of frames.
- Various sampling rates(range(0.5,2)) on a video snippet to obtain snippets with different playing **speed**

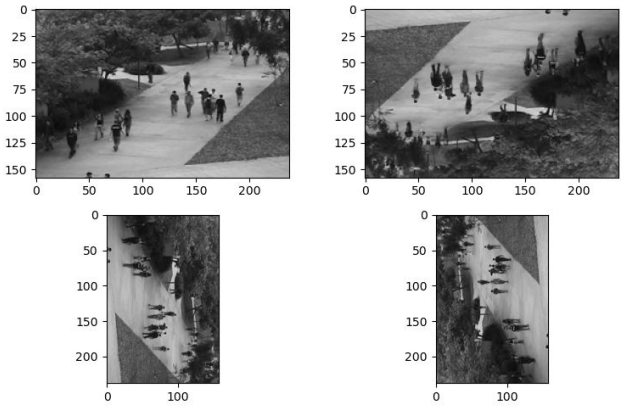


Fig 3: Jittered rotation augmentation

Output Dim: 25,8,3,158,238

We have considered 8 frames as one snippet. One of the video folders consists of 200 frames. 200/8 = 25 snippets

Reference : Huang, Chao, et al. "Abnormal event detection using deep contrastive learning for intelligent video surveillance system." *IEEE Transactions on Industrial Informatics* 18.8 (2021): 5171-5179.

Implementation



Pre-processing :

- The input videos are preprocessed to with the resolution of 256 × 256 and the intensity of [−1, 1] for all datasets.
- From a Given snippets,to obtain snippet level features Encoder was used.
- 3D-ResNets were used as an encoder. 3-D convolution kernel is adopted to capture the spatio temporal patterns of normal events.
- These original and augmented snippets are then fed into an encoder, which extracts snippet-level features for the original snippet, $f(S_i)$, and the augmented snippets, $\{f(S_i^{T1}),\ldots,f(S_i^{T4})$.
- **Output Dim : 25,512,1,32,32**

25 snippets,8 frames=200 frames

Blocks	Architecture	Output $T \times S^2$
conv	$7 \times 7^2, 64$ stride:$1,1^2$ +ReLu+BN	8×256^2
res1	$\begin{bmatrix} 3 \times 3^2, 128 & \text{stride} : 2, 2^2 \\ 3 \times 3^2, 128 & \text{stride} : 1, 1^2 \end{bmatrix} \times 1$ $\begin{bmatrix} 3 \times 3^2, 128 & \text{stride} : 1, 1^2 \\ 3 \times 3^2, 128 & \text{stride} : 1, 1^2 \end{bmatrix} \times 2$	4×128^2
res2	$\begin{bmatrix} 3 \times 3^2, 256 & \text{stride} : 2, 2^2 \\ 3 \times 3^2, 256 & \text{stride} : 1, 1^2 \end{bmatrix} \times 1$ $\begin{bmatrix} 3 \times 3^2, 256 & \text{stride} : 1, 1^2 \\ 3 \times 3^2, 256 & \text{stride} : 1, 1^2 \end{bmatrix} \times 3$	2×64^2
res3	$\begin{bmatrix} 3 \times 3^2, 512 & \text{stride} : 2, 2^2 \\ 3 \times 3^2, 512 & \text{stride} : 1, 1^2 \end{bmatrix} \times 1$ $\begin{bmatrix} 3 \times 3^2, 512 & \text{stride} : 1, 1^2 \\ 3 \times 3^2, 512 & \text{stride} : 1, 1^2 \end{bmatrix} \times 5$	1×32^2

Fig 4: Network Architecture of 3D Encoder

Reference : Huang, Chao, et al. "Abnormal event detection using deep contrastive learning for intelligent video surveillance system." *IEEE Transactions on Industrial Informatics* 18.8 (2021): 5171-5179.

Implementation



Frame Prediction :

- As for the raw videos, we feed the raw snippet-level features ($f(S_i)$) into the prediction task head to conduct frame prediction task.In the prediction head,we incorporate decoder composed of three upsampling layers and three residual blocks with 2-D convolutions. Each upsampling operation adopts nearest neighbor interpolation. The last convolution layer in the decoder has only three filters to reconstruct the RGB frame.
- Frame prediction was done by taking 7 frames and **predicting the 8th frame** in a snippet, rather than reconstructing all frames, where during inference it doesn't have an impact as shown in Fig 5.
- When an **abnormal behavior** occurs, such as fighting, the corresponding input video snippet usually cannot contain complete spatio-temporal information for the model of normal patterns to accurately predict frame, so as to the anomaly can be detected.
- **Loss computation**, Let h_{Tpre} is the frame prediction task head. We apply intensity and gradient constraints (L_{int} and L_{grd})on the frame prediction task.

$$L_{Tpre} = L_{int} + L_{grd}$$

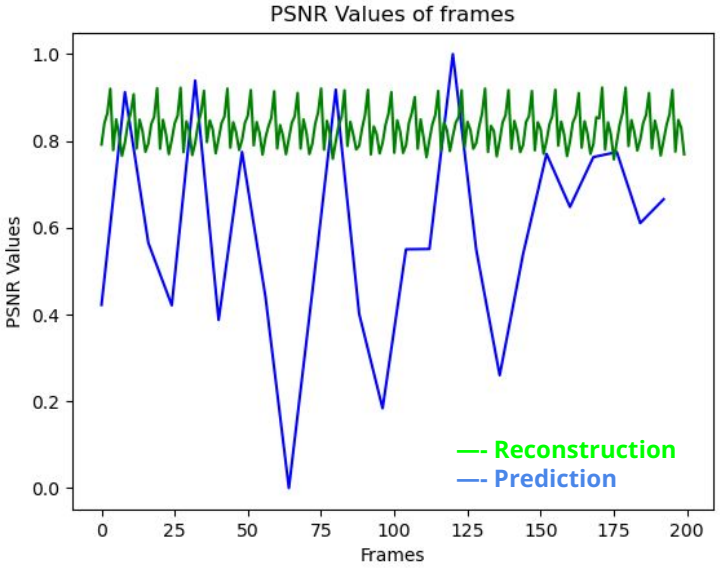


Fig 5: Comparison of PSNR values through Frame Reconstruction and Prediction.

Reference : Huang, Chao, et al. "Abnormal event detection using deep contrastive learning for intelligent video surveillance system." *IEEE Transactions on Industrial Informatics* 18.8 (2021): 5171-5179.

Implementation



Self Supervised Learning Task :

- We create a pretext task for each type of augmentations.For solving these self-supervised tasks, we apply multiple task heads h_{Tn} ,which take the snippet-level features $f(S_{Tn})$ as the input.
- Self-supervised task was considered as **classification problem**.
- For all self-supervised tasks,considered as equally important for incorporating temporal context information
- The groundtruth labels for 4 tasks are:
  - Temporal reverse : 2 i.e, Snippet is reversed or not
  - Temporal speed : 2 i.e, Frames are accelerated or consecutive frames
  - Temporal shuffle : 2 i.e, Frames in snippet shuffled or not
  - Jittered Rotation : 4 i.e, 4 types of shuffle (0,90,180,270)
- Loss computation was done at a time by combining all 4 task heads losses.
- We utilize the **cross entropy loss** to learn each self-supervised task.
- Network architecture composed of a fully connected layer, a reshaping operation, and a linear layer.
- For optimization **Adam** was used.

Reference : D. Wei, J. J. Lim, A. Zisserman, and W. T. Freeman, “Learning and using the arrow of time,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2018, pp. 8052–8060.

Implementation



Contrastive Learning Task :

- Let p denotes a projection head that projects features f(S) (denoted by f_S) to embedding space for performing contrastive learning.A linear layer is used to acquire $z_S = p(f_S) = W^{(1)} f_S$,followed by a l2 normalization layer. Multi-layer perceptron (MLP) is also applicable here. In this case, $z_S = p(f_S) = W^{(2)} \sigma (W^{(1)} f_S)$ where σ is an rectified linear unit (ReLU). The features of raw snippet Si and its augmented snippets STn i are considered as positive pairs, while otherwise are regarded as negative sample pairs.
- For Loss Computation **InfoNCE** loss was used to force the positive pairs to be similar with each other and encourage the negative pairs to be dissimilar.
- Network Architecture was composed of a fully connected layer, a l2 normalized layer and a linear layer based on **MoCo-v2** Framework.
- For Optimization, SGD was used.

Reference : K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, “Momentum contrast for unsupervised visual representation learning,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2020, pp. 9729–9738

Implementation



Inference :

During inference, for each test sample, we calculate the final regularity score of it as the average score given by multiple task heads.

- For **Frame prediction(S_{Tpre})**,regularity score is error between prediction frame Y and its corresponding ground-truth Y with peak signal to noise ratio (PSNR).
- For the **self-supervised tasks(S_{Ts})**, we regard the average probability of that video snippets are augmented with the corresponding transformations as anomaly scores.
- For the **contrastive learning task(S_{Tcon})**, we take the total contrast similarities between features of different augmented snippets as the regularity score.
- The regularity score is smoothed by a temporal Gaussian filter.

$$S=S_{Tpre}+S_{Tst}+S_{Tcon}$$

Reference : Huang, Chao, et al. "Abnormal event detection using deep contrastive learning for intelligent video surveillance system." *IEEE Transactions on Industrial Informatics* 18.8 (2021): 5171-5179.

Implementation

Results:



The Regularity Scores of below is from one of the testing video folders in UCSD Ped1 dataset. Higher Regularity score means normal and low indicates abnormal activity.The Comparison of regularity scores and frames makes to believe that anomaly detection was done successfully with **training of one video and testing with one video.**

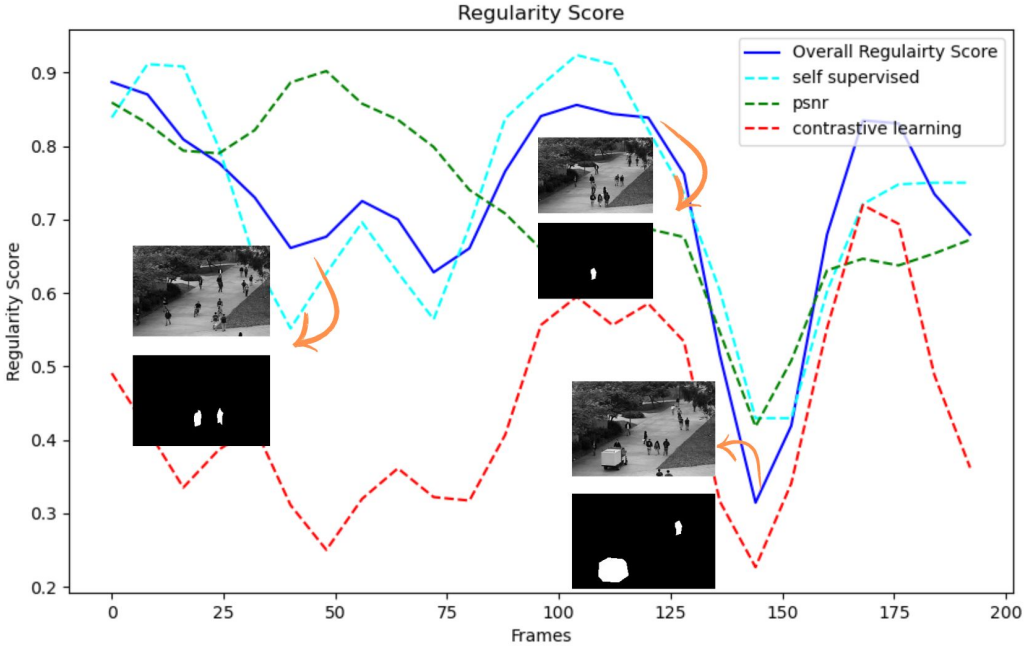


Fig 6 : Video Anomaly Detection with multi task approach.

Implementation



Results:

Initially, Self Supervised task wasn't include in the approach but through experimentation with self supervised task, robustness of detection was increased.

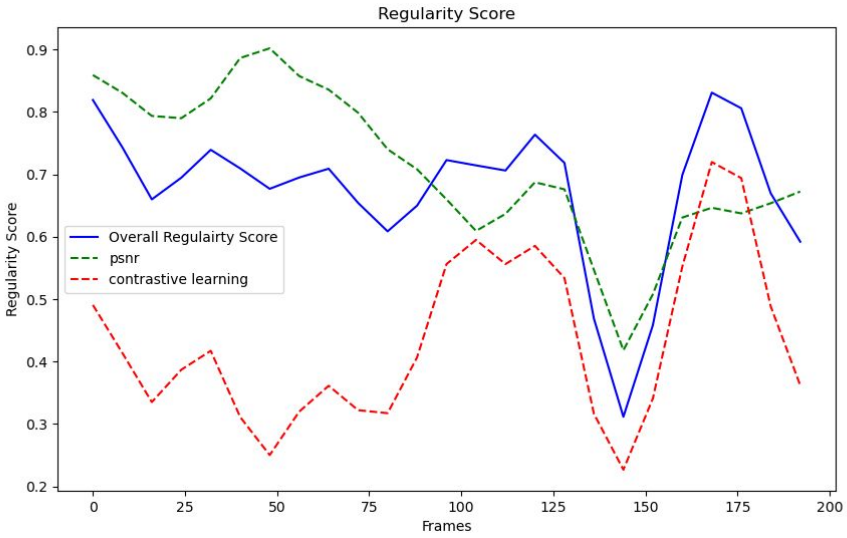


Fig 7 : Video Anomaly Detection without self supervised task.

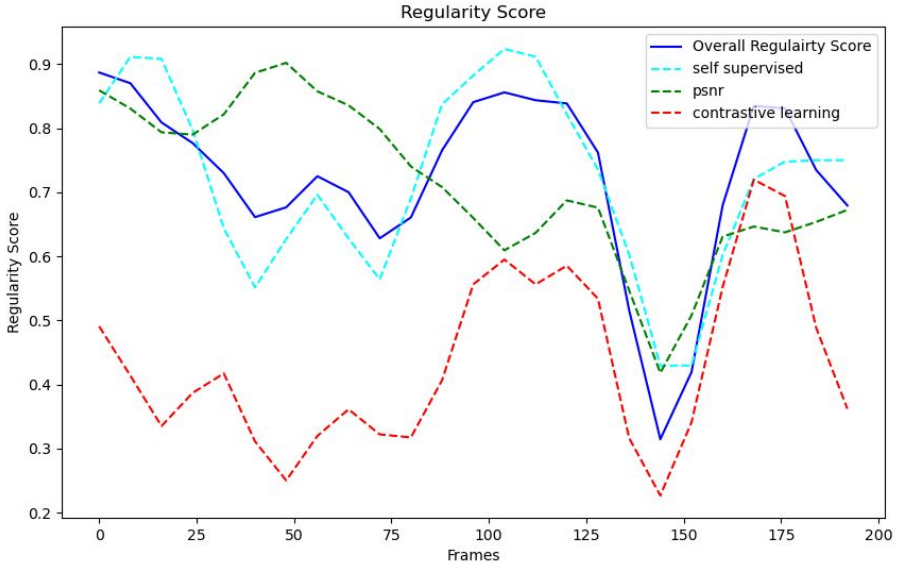
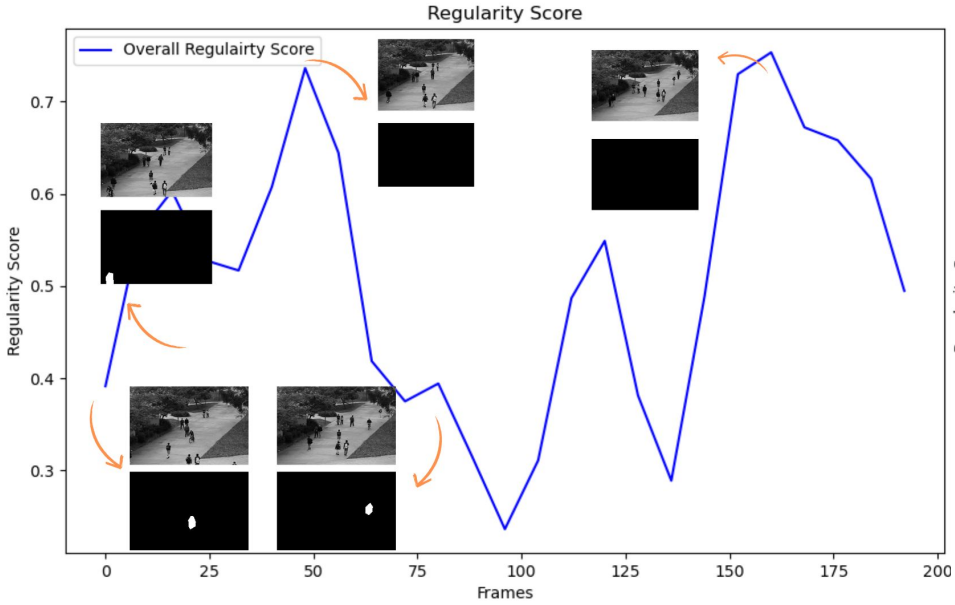


Fig 8 : Video Anomaly Detection with self supervised task.

Implementation



Results:



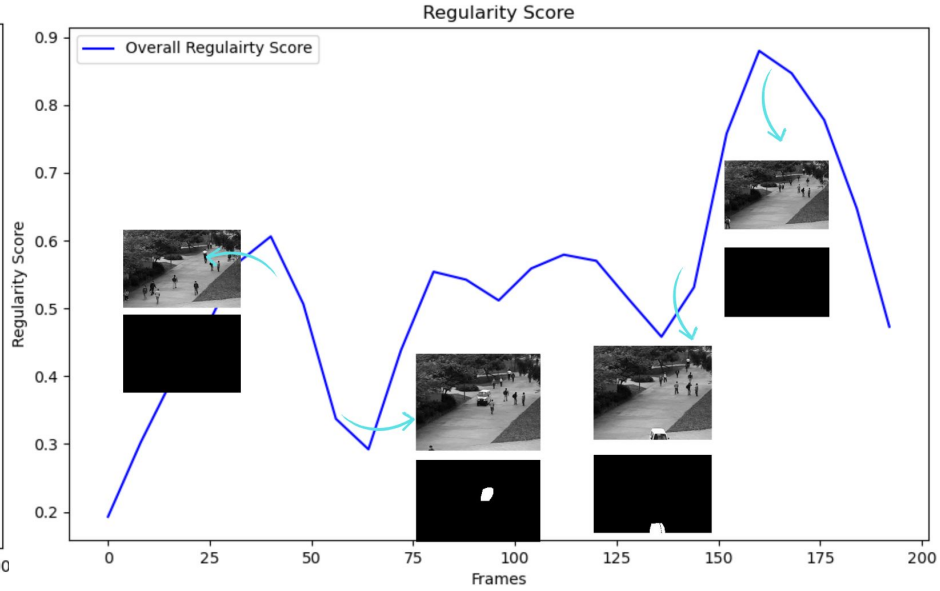


Fig 9 : Regularity scores of testing videos from UCSD Pred1 dataset

Learnings



By doing this project the following were learned

- Implementation in Pytorch framework.
- Writing code from scratch,using frameworks from paper as a reference.
- Design of model architecture according to the problem.
- Multi Task learning implementation.
- Dealing with Loss computations.
- Suitable Optimization techniques implementation according to the framework.
- Implementation of Normalization techniques for images,batches and features.
- Team Management - Dividing of model blocks among team members

Future Scope



For more accurate detection.

- Creation of overlapping video snippets
- Training with more videos than one video.

Individual Contributions



"Alone we can do so little; together we can do so much." – Helen Keller

Inspired by the wisdom of above quote, we have divided our work equally among team members as follows:

- **Dhana Lakshmi**: Preprocessing task
- **Sai Kiran K**: Self-supervised tasks and integration of all tasks
- **Jayadeep**: Contrastive learning task
- **Beeram Sandya**: Data augmentation task
- **Cheekatla Hema Sri**: Frame prediction task

The literature survey and code implementation were carried out individually by each member based on their assigned tasks.

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